

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with a strong foundation in probability, statistical inference, and computational analysis. Builds and benchmarks quantitative research code in R, C++, and Python, with experience translating models into efficient implementations, validating numerical correctness, and working on data-intensive problems with clear, structured reasoning.

EDUCATION

Indiana University, College of Arts and Sciences

Bloomington, IN

Master of Science in Statistical Science

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II

Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Dhaka, Bangladesh

Bachelor of Science in Physics

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Implemented core Bayesian brain-mapping routines in C++ using Rcpp and RcppArmadillo for high-dimensional neuroimaging data.
- Reproduced the original R package outputs and validated the new implementation through unit testing and runtime benchmarking.
- Delivered a 4 to 5× speedup in core computations by moving numerically intensive work into compiled C++ code.
- Built reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.

Teaching Assistant, Department of Statistics

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Support undergraduate statistics courses through instruction, discussion sections, and individualized problem-solving help.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen conceptual understanding and quantitative communication.

Senior Mathematics Instructor

2016 – 2025

Udvash Academic & Admission Care

Dhaka, Bangladesh

- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Prepared students for competitive university admission examinations using structured lessons, practice materials, and academic planning.
- Translated advanced mathematical ideas into clear problem-solving strategies for students with varied backgrounds.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct research on Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Matthew T. Pratola.
- Apply statistical modeling and reproducible analysis workflows to evaluate predictive performance on time-series data.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Developed an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation and transparent statistical computation.
- Validated outputs with numerical checks and clear visualization of model fit and residual behavior.

TECHNICAL SKILLS

Quantitative Methods: Probability, statistical inference, Bayesian modeling, BART, MCMC, hierarchical modeling, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, experimental design, model selection, and predictive model evaluation

Programming: R, Python, C, C++, MATLAB, Bash, HTML, CSS, and JavaScript

Computing and Tools: Numerical optimization, algorithm benchmarking, code profiling, unit testing, Git, GitHub, LaTeX, RStudio, and VS Code

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online linear algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.